A MINI PROJECT REPORT

ON

“ THE COVID STATE WISE VACCINE ”

SUBMITTED TOWARDS THE
PARTIAL FULFILLMENT OF THE REQUIREMENTS OF

BACHELOR OF ENGINEERING (TE Computer Engineering)

BY

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Under The Guidance of
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“ येथे बहुतांचे हित ”
MMIT

“Towards Ubiquitous Computing Technology”

DEPARTMENT OF COMPUTER ENGINEERING
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Lohgaon, Pune- 411 047
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“Towards Ubiquitous Computing Technology”
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CERTIFICATE

This is to certify that the Project Entitled
“The Covid State wise Vaccine ”

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is a bonafide work carried out by students under the supervision of Prof. Devyani Bonde and it is submitted towards the partial fulfillment of the requirement of Bachelor of Engineering (TE Computer Engineering) Mini Project.

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Abstract

This mini-project focuses on analyzing the COVID-19 vaccination data in India sourced from the dataset available on Kaggle. The dataset provides insights into the vaccination drive across different states in India, detailing the number of individuals vaccinated with the first and second doses, as well as demographic information such as gender.

The project is divided into 3 parts analyzing the Vaccination process state wise and gender wise for dose 1 and 2.The State wise First Dose Vaccination analysis identifies the total number of individuals vaccinated with the first dose in each state of India. It provides insights into the distribution and progress of the vaccination drive across different regions.While, the State Wise Second Dose Vaccination analysis calculates the total number of individuals vaccinated with the second dose in each state. Understanding the uptake of the second dose is crucial for assessing the completion of the vaccination process and ensuring sufficient immunity against COVID-19.

The project also explores the gender distribution of vaccinated individuals in India. By analyzing the number of males and females vaccinated, it sheds light on any potential gender disparities in access to vaccination and highlights areas for targeted interventions.The findings of this analysis provide valuable insights for policymakers, healthcare professionals, and researchers involved in the COVID-19 vaccination efforts in India. Understanding the vaccination coverage across different states and demographic groups is essential for optimizing resource allocation, addressing inequities, and ensuring a comprehensive approach to combating the pandemic.

Thus,the analysis of COVID-19 vaccination data in India underscores the importance of data-driven decision-making in optimizing vaccination strategies and ensuring equitable access to vaccination services. This analysis serves as a valuable tool for guiding vaccination policies, resource allocation, and outreach efforts, ultimately contributing to the control of COVID-19 in India

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INTRODUCTION

The emergence of the COVID-19 pandemic has presented unparalleled challenges to global health systems, necessitating swift and effective responses to mitigate the spread of the virus and protect public health. As one of the most populous countries in the world, India faces a monumental task in managing the impact of the pandemic and implementing strategies to control transmission. Central to these efforts is the deployment of vaccines to achieve widespread immunity and curb the spread of the virus.

The Government of India launched an ambitious vaccination campaign, aiming to inoculate its vast population against COVID-19. Since the rollout of vaccines, monitoring and analyzing vaccination data have become imperative for evaluating the effectiveness of the vaccination drive, identifying areas of success, and pinpointing challenges that need to be addressed. In this context, leveraging data analytics techniques to delve into the intricacies of vaccination trends becomes crucial for informed decision-making and resource allocation.

This project delves into the realm of COVID-19 vaccination data in India, utilizing a comprehensive dataset sourced from Kaggle. This dataset provides granular insights into the vaccination landscape across various states and territories, encompassing crucial information such as the number of doses administered, demographic breakdowns, and geographical distribution. By harnessing the power of data analytics, we endeavor to unravel the multifaceted dimensions of India's vaccination campaign, shedding light on key trends and patterns that can inform policy formulation and intervention strategies.

At the heart of our analysis lies the exploration of state-wise vaccination coverage. By scrutinizing vaccination rates across different regions of India, we aim to discern disparities in coverage levels, identify regions with robust vaccination uptake, and pinpoint areas requiring intensified efforts and targeted interventions. Additionally, we delve into the uptake of the second dose of the vaccine, a critical metric for assessing the completion of the vaccination process and ensuring optimal immunity levels within the population.

Furthermore, our analysis extends to examining gender disparities in vaccination uptake. By dissecting the gender distribution of vaccinated individuals, we seek to uncover any inequities in access to vaccination services, thereby advocating for gender-inclusive strategies to ensure equitable vaccine distribution and coverage.

Through this project, we aspire to provide actionable insights to stakeholders involved in India's vaccination drive, including policymakers, healthcare professionals, and public health authorities. By leveraging data-driven insights, we aim to bolster vaccination efforts, optimize resource allocation, and contribute to the overarching goal of combating the COVID-19 pandemic in India.

OBJECTIVE

The primary objective of this project is to conduct a comprehensive analysis of COVID-19 vaccination data in India, with a focus on understanding vaccination trends, identifying challenges, and proposing data-driven solutions to optimize vaccination efforts. Through meticulous exploration of the dataset and application of advanced analytics techniques, we aim to achieve the following objectives:

1. **Evaluate State-wise Vaccination Coverage:**
   Our first objective is to assess the distribution of vaccination coverage across different states and territories in India. By analyzing the number of doses administered and vaccination rates in each region, we seek to identify areas with high vaccination uptake as well as regions that may require targeted interventions to enhance coverage.
2. **Assess Second Dose Uptake:**
   Building upon the analysis of first dose vaccination coverage, we aim to delve deeper into the uptake of the second dose of the COVID-19 vaccine. Understanding the proportion of individuals who have completed the vaccination regimen is essential for ensuring optimal immunity levels and assessing the success of follow-up efforts.
3. **Examine Gender Disparities in Vaccination Uptake:**
   Gender equity in vaccination access is crucial for ensuring equitable distribution of vaccines and achieving herd immunity. Therefore, our objective is to analyze the gender distribution of vaccinated individuals and identify any disparities in vaccination uptake between males and females. This analysis will inform targeted interventions to address gender-specific barriers and promote inclusive vaccination coverage.
4. **Provide Insights for Policy Formulation and Decision-Making:**
   A key objective of this project is to generate actionable insights that can inform policy formulation, resource allocation, and decision-making processes related to India's COVID-19 vaccination campaign. By presenting data-driven recommendations, we aim to support policymakers, healthcare professionals, and public health authorities in optimizing vaccination strategies and overcoming challenges.
5. **Foster Transparency and Public Trust:**
   By conducting a rigorous and transparent analysis of vaccination data, we aim to foster public trust and confidence in the vaccination process. Transparent communication of findings and recommendations will promote accountability.

PROBLEM STATEMENT

Use the following covid_vaccine_statewise.csv dataset and perform following analytics on the given dataset
https://www.kaggle.com/sudalairajkumar/covid19-in-india?select=covid_vaccine_statewise. csv

1. Describe the dataset
2. Number of persons state wise vaccinated for first dose in India
3. Number of persons state wise vaccinated for second dose in India.
4. Number of Males vaccinated
5. Number of females vaccinated

LITERATURE SURVEY

The coronavirus (SARS-CoV-2) manifested in the Hubei province (Wuhan) of China sometime in October 2019. In a short span of three months, the virus had spread to over 100 countries with more than 2 lakh cases reported globally, resulting in the World Health Organization (WHO) declaring it a pandemic. The scale of transmission, infections and mortality overwhelmed the healthcare systems, leading to countries enforcing complete lockdowns to stop the spread of the coronavirus. In the absence of vaccines, the strategy followed by governments all over the world to reduce the risk of contagion was non-pharmaceutical interventions (NPIs), such as enforcing restrictions on movements, the use of facial masks, mandatory quarantines, physical distancing, travel restrictions and lockdowns. We are presented with an analytical framework to probe whether vaccine hesitancy, socioeconomic factors and multi-dimensional deprivations (MPI) play a role in determining COVID-19 vaccination uptake. Our exploratory analysis reveals that COVID-19 vaccine hesitancy has a negative and statistically significant impact on COVID-19 vaccination coverage.We found that, as males' access to the internet increases, vaccination coverage also increases.This may be attributed to India's reliance on digital tools (COWIN, AAROGYA SETU, Imphal, India) to allocate and register for COVID-19 vaccines and the associated digital divide (males have greater digital excess than females). Conversely, females' access to the internet is statistically significant and inversely associated with coverage. This can be attributed to higher vaccine hesitancy among the female population and lower utilization of health services by females.

The adverse impacts of the COVID-19 pandemic have been felt in all spheres of human life, with unprecedented challenges to public health and the economy. Previously, a study was designed to explore the factors associated with COVID-19 vaccination coverage in India at the district level.The study used data from COVID-19 vaccination in India combined with several other administrative data to create a unique data set that facilitated a spatio–temporal exploratory analysis by uncovering the factors associated with vaccination rates across different vaccination phases and districts. They found evidence that past reported infection rates were positively correlated with COVID-19 vaccination outcomes. Past cumulative COVID-19 deaths as a proportion of district populations were associated with lower COVID-19 vaccination, but the percentage of past reported infection was positively correlated with first-dose COVID-19 vaccination, which might indicate a positive role of higher awareness created by a higher reported infection rate. Districts that on average had a higher population burden per health center were likely to have lower COVID-19 vaccination rates. Vaccination rates were lower in rural areas relative to urban areas, whereas the association with literacy rate was positive. Districts with a higher percentage of children with complete immunization were associated with higher COVID-19 vaccination, whereas low vaccination was observed in districts that had higher percentages of wasted children. COVID-19 vaccination was lower among pregnant and lactating women. Higher vaccination was observed among populations with higher blood pressure and hypertension

REQUIREMENT SPECIFICATION

- **Hardware Specification**
  1. Laptop or PC

- **Software Specification**
  1. Operating System : Windows 7,8,10/11.
  2. Browser : Google Chrome or Firefox
  3. Programming Language : Python
  4. IDE : Colab.
  5. Libraries : Numpy,Seaborn, Matplotlib and Pandas .

Implementation

- **Code :-**

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load the dataset
df = pd.read_csv('data/covid_vaccine_statewise.csv')

# a. Describe the dataset
print("a. Dataset Description:")
print(df.describe())
print("\nDataset Info:")
print(df.info())
print("\nFirst 5 rows:")
print(df.head())

# b. Number of persons state wise vaccinated for first dose in India
print("\n\nb. Number of persons state-wise vaccinated for first dose:")
first_dose = df.groupby('State')['First Dose Administered'].sum().sort_values(ascending=False)
print(first_dose)

# Plot for first dose
plt.figure(figsize=(12, 8))
first_dose.head(10).plot(kind='bar')
plt.title('Top 10 States: First Dose Vaccinations')
plt.xlabel('State')
plt.ylabel('Number of Persons')
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig('first_dose_statewise.png')
# plt.show()

# c. Number of persons state wise vaccinated for second dose in India
print("\n\nc. Number of persons state-wise vaccinated for second dose:")
second_dose = df.groupby('State')['Second Dose Administered'].sum().sort_values(ascending=False)
print(second_dose)

# Plot for second dose
plt.figure(figsize=(12, 8))
second_dose.head(10).plot(kind='bar')
```

```
plt.title('Top 10 States: Second Dose Vaccinations')
plt.xlabel('State')
plt.ylabel('Number of Persons')
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig('second_dose_statewise.png')
# plt.show()

# d. Number of males vaccinated
print("\n\nd. Total number of males vaccinated (doses administered):")
males_vaccinated = df['Male (Doses Administered)'].sum()
print(f"Total doses administered to males: {males_vaccinated}")

# e. Number of females vaccinated
print("\n\ne. Total number of females vaccinated (doses administered):")
females_vaccinated = df['Female (Doses Administered)'].sum()
print(f"Total doses administered to females: {females_vaccinated}")

# Gender comparison plot
gender_data = {'Males': males_vaccinated, 'Females': females_vaccinated}
plt.figure(figsize=(8, 6))
plt.bar(gender_data.keys(), gender_data.values())
plt.title('Total Doses Administered by Gender')
plt.xlabel('Gender')
plt.ylabel('Number of Doses')
plt.tight_layout()
plt.savefig('gender_vaccinations.png')
# plt.show()
print("\nPlots saved as PNG files in the project directory.")
```

- **Output :-**

```
PS C:\Users\SANKET PATIL\OneDrive\Desktop\covid 19> python scripts/covid_analysis.py
a. Dataset Description:
       Total Doses Administered  ...  Total Individuals Vaccinated
count              7.621000e+03  ...                  5.919000e+03
mean               9.188171e+06  ...                  4.547842e+06
std                3.746180e+07  ...                  1.834182e+07
min                7.000000e+00  ...                  7.000000e+00
25%                1.356570e+05  ...                  7.427550e+04
50%                8.182020e+05  ...                  4.022880e+05
75%                6.625243e+06  ...                  3.501562e+06
max                5.132284e+08  ...                  2.506569e+08
```

```
[8 rows x 22 columns]

Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7845 entries, 0 to 7844
Data columns (total 24 columns):
 #   Column                               Non-Null Count  Dtype
---  -------                              --------------  -----
 0   Updated On                           7845 non-null   object
 1   State                                7845 non-null   object
 2   Total Doses Administered             7621 non-null   float64
 3   Sessions                             7621 non-null   float64
 4   Sites                                7621 non-null   float64
 5   First Dose Administered              7621 non-null   float64
 6   Second Dose Administered             7621 non-null   float64
 7   Male (Doses Administered)            7461 non-null   float64
 8   Female (Doses Administered)          7461 non-null   float64
 9   Transgender (Doses Administered)     7461 non-null   float64
 10  Covaxin (Doses Administered)         7621 non-null   float64
 11  CoviShield (Doses Administered)      7621 non-null   float64
 12  Sputnik V (Doses Administered)       2995 non-null   float64
 13  AEFI                                 5438 non-null   float64
 14  18-44 Years (Doses Administered)     1702 non-null   float64
 15  45-60 Years (Doses Administered)     1702 non-null   float64
 16  60+ Years (Doses Administered)       1702 non-null   float64
 17  18-44 Years(Individuals Vaccinated)  3733 non-null   float64
 18  45-60 Years(Individuals Vaccinated)  3734 non-null   float64
 19  60+ Years(Individuals Vaccinated)    3734 non-null   float64
 20  Male(Individuals Vaccinated)         160 non-null    float64
 21  Female(Individuals Vaccinated)       160 non-null    float64
 22  Transgender(Individuals Vaccinated)  160 non-null    float64
 23  Total Individuals Vaccinated         5919 non-null   float64
dtypes: float64(22), object(2)
memory usage: 1.4+ MB
None

First 5 rows:
  Updated On  ... Total Individuals Vaccinated
0 16/01/2021  ...                      48276.0
1 17/01/2021  ...                      58604.0
2 18/01/2021  ...                      99449.0
3 19/01/2021  ...                     195525.0
4 20/01/2021  ...                     251280.0

[5 rows x 24 columns]
```

b. Number of persons state-wise vaccinated for first dose:

```
State
India                          2.826214e+10
Uttar Pradesh                  2.788411e+09
Maharashtra                    2.784364e+09
Rajasthan                      2.201044e+09
Gujarat                        2.131646e+09
Karnataka                      1.873330e+09
Madhya Pradesh                 1.796605e+09
West Bengal                    1.796450e+09
Bihar                          1.470503e+09
Tamil Nadu                     1.288533e+09
Andhra Pradesh                 1.232861e+09
Kerala                         1.193845e+09
Odisha                         1.032633e+09
Telangana                      8.803206e+08
Chhattisgarh                   7.960029e+08
Haryana                        7.557984e+08
Delhi                          6.243395e+08
Jharkhand                      6.036737e+08
Assam                          5.856002e+08
Punjab                         5.843466e+08
Jammu and Kashmir              4.101018e+08
Uttarakhand                    3.631914e+08
Himachal Pradesh               3.162940e+08
Tripura                        1.926897e+08
Goa                            7.599137e+07
Manipur                        6.740957e+07
Meghalaya                      6.261597e+07
Arunachal Pradesh              4.900498e+07
Mizoram                        4.787308e+07
Chandigarh                     4.470310e+07
Nagaland                       4.241077e+07
Puducherry                     4.134686e+07
Sikkim                         3.698093e+07
Dadra and Nagar Haveli and Daman and Diu    3.359506e+07
Ladakh                         1.780925e+07
Andaman and Nicobar Islands    1.642585e+07
Lakshadweep                    4.363655e+06
Name: First Dose Administered, dtype: float64
```

c. Number of persons state-wise vaccinated for second dose:

State	
India	6.759621e+09
Maharashtra	7.128811e+08
Gujarat	6.004184e+08
West Bengal	5.861469e+08
Uttar Pradesh	5.544351e+08
Rajasthan	4.917030e+08
Karnataka	4.271872e+08
Kerala	3.640488e+08
Andhra Pradesh	3.588176e+08
Madhya Pradesh	3.169330e+08
Tamil Nadu	2.906706e+08
Bihar	2.707906e+08
Odisha	2.513028e+08
Telangana	1.981529e+08
Delhi	1.882189e+08
Chhattisgarh	1.721204e+08
Haryana	1.586561e+08
Assam	1.307888e+08
Jharkhand	1.221211e+08
Punjab	1.211210e+08
Uttarakhand	1.000850e+08
Jammu and Kashmir	8.595165e+07
Himachal Pradesh	7.383858e+07
Tripura	6.527014e+07
Goa	1.619817e+07
Meghalaya	1.216663e+07
Arunachal Pradesh	1.193232e+07
Manipur	1.185815e+07
Chandigarh	1.159374e+07
Mizoram	9.998418e+06
Sikkim	9.723640e+06
Nagaland	9.204637e+06
Puducherry	8.608859e+06
Ladakh	5.453762e+06
Dadra and Nagar Haveli and Daman and Diu	4.594416e+06
Andaman and Nicobar Islands	4.118554e+06
Lakshadweep	1.056446e+06

Name: Second Dose Administered, dtype: float64

d. Total number of males vaccinated (doses administered):
Total doses administered to males: 27009983996.0

e. Total number of females vaccinated (doses administered):
Total doses administered to females: 23639554465.0

Plots saved as PNG files in the project directory.
PS C:\Users\SANKET PATIL\OneDrive\Desktop\covid 19>

Fig 1 : Box Plot For Dose 1

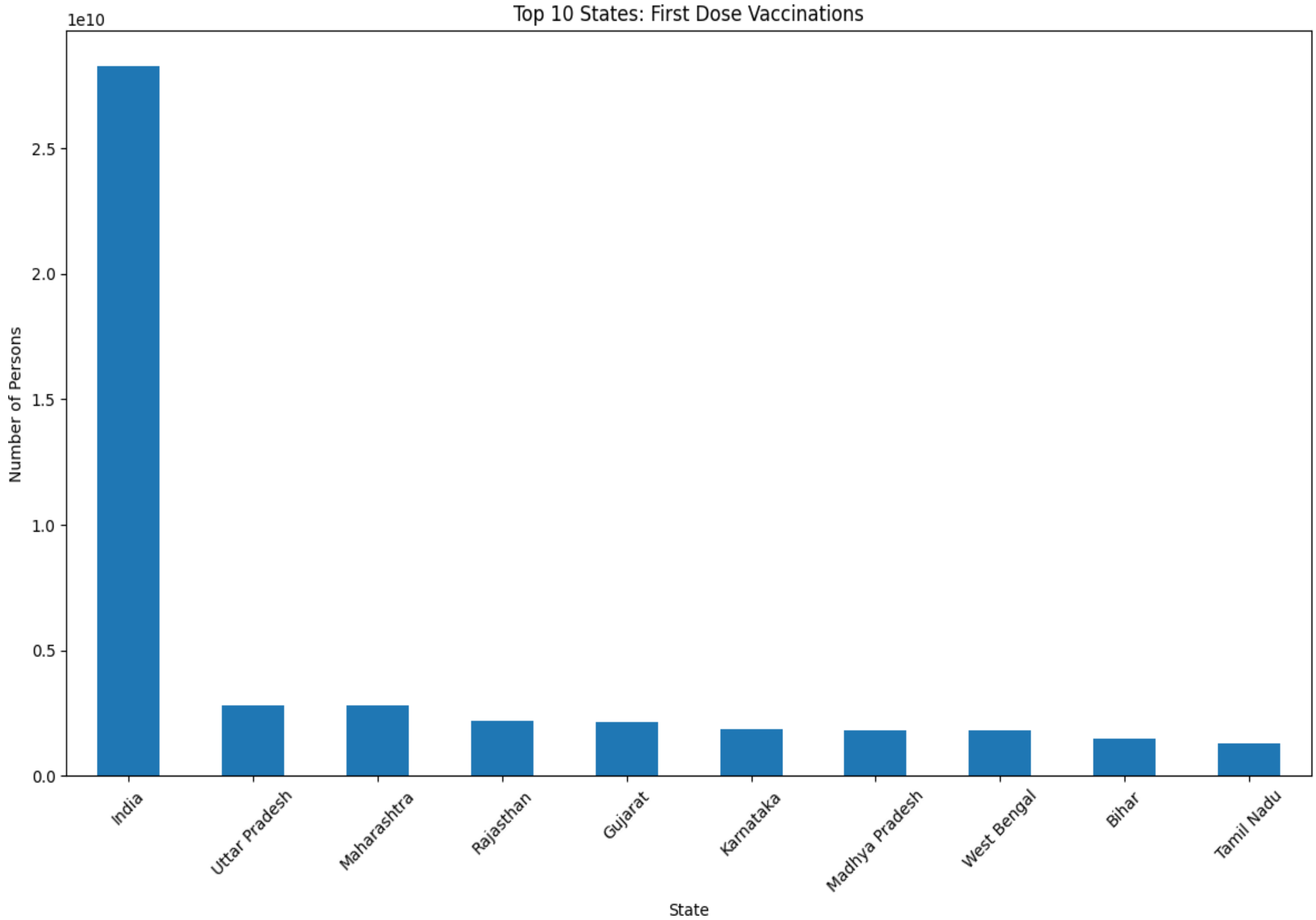


Fig 2 : Box Plot For Dose 2

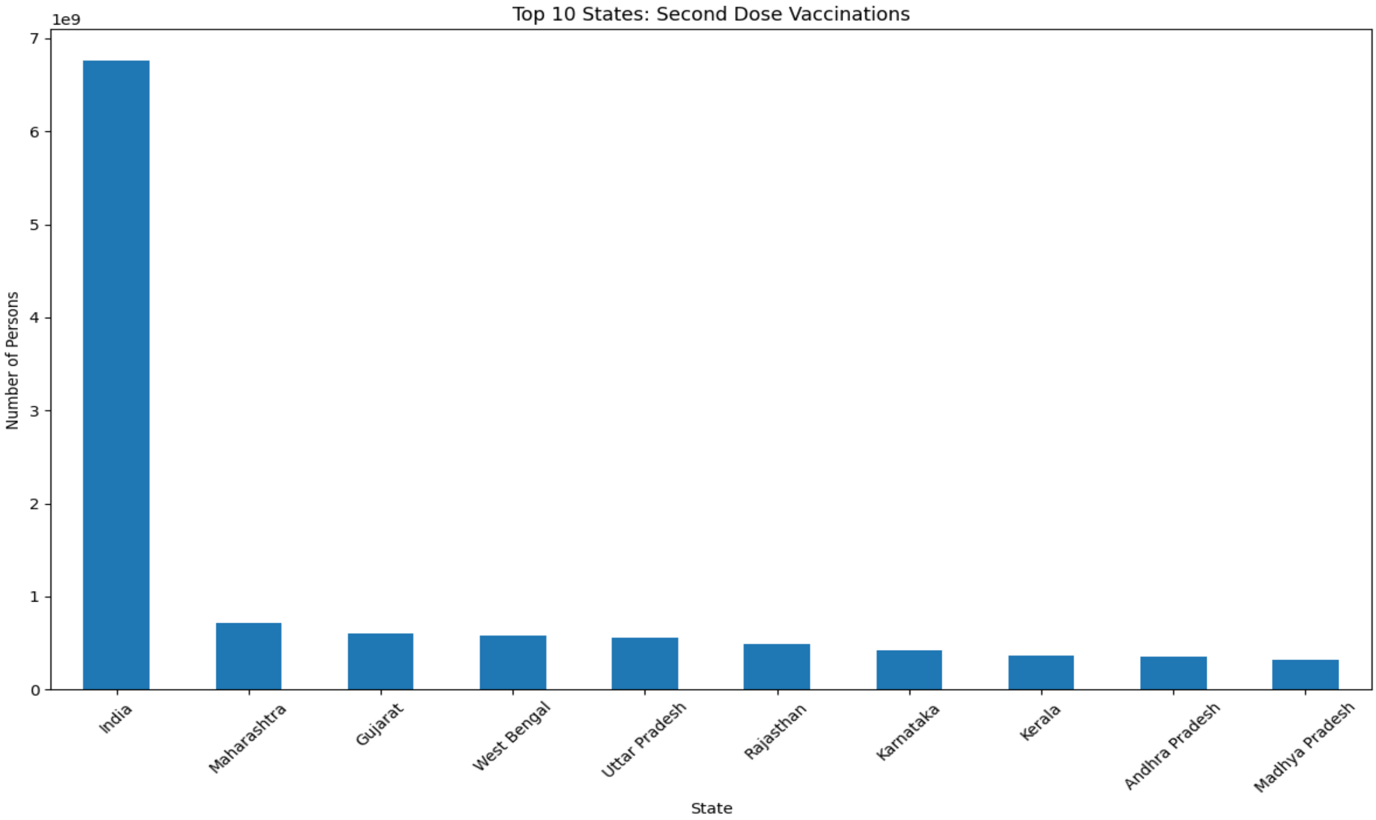
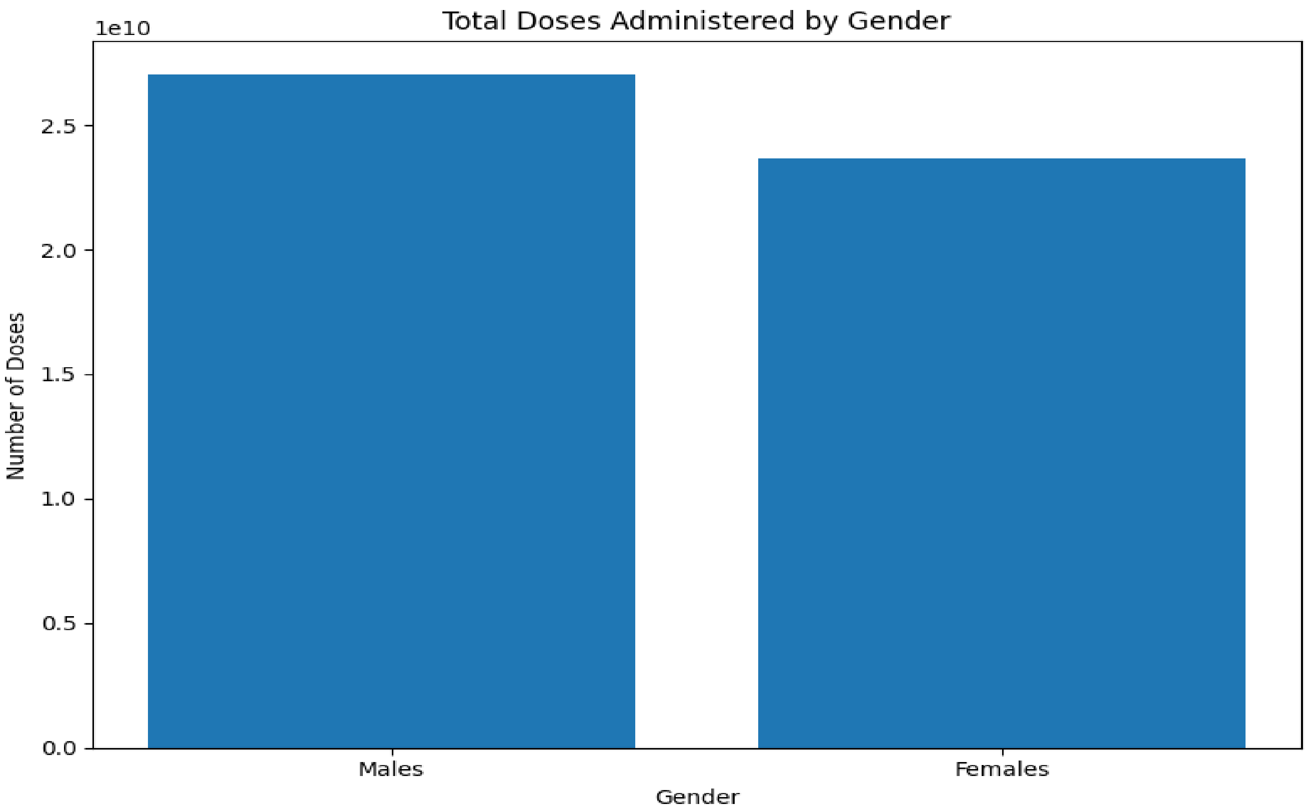


Fig 3 : Gender Vaccination



Conclusion

Thus, the COVID-19 Vaccination Data Analysis project has provided valuable insights into the vaccination trends and strategies in India, shedding light on key aspects of the vaccination drive and offering actionable recommendations for policymakers, healthcare professionals, and researchers.

Through rigorous data analysis and visualization, we have examined state-wise vaccination coverage, second dose uptake, and gender distribution of vaccinated individuals. The analysis revealed variations in vaccination coverage among different states, emphasizing the need for targeted interventions to address disparities and improve overall coverage. Additionally, insights into second dose uptake highlighted the importance of completing the vaccination regimen for achieving optimal immunity levels and controlling the spread of the virus.

Furthermore, the examination of gender distribution in vaccination uptake uncovered potential disparities in access to vaccination services, underscoring the importance of promoting equitable access and addressing barriers to vaccine acceptance.

Therefore, the COVID-19 Vaccination Data Analysis project serves as a valuable tool for informing evidence-based decision-making, resource allocation, and targeted interventions to combat the COVID-19 pandemic in India. By leveraging data-driven insights, stakeholders can optimize vaccination strategies, address inequities, and ultimately contribute to controlling the spread of the virus and protecting public health. Moving forward, continued monitoring and analysis of vaccination data will be essential for adapting strategies and ensuring the success of India's vaccination campaign.